KEDGE Business School EBP1 Quantitative Methods 1

Worksheet 4 — Bivariate distributions and regression

Exercise statements

Badr ABOUFARASSE, PhD — 2026–2027

1 Core applications

Master the essential definitions and calculations from the course.

Core application — 15 min

1.1 Exercise 3 – Mayer line

For $(1,7),(2,10),(3,10),(4,14),(5,14)$, form two sub-clouds, calculate mean points A and B and determine the Mayer line.

2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice — 35 min

2.1 Exercise 1 – Complete linear regression

Consider $(1,3),(2,5),(3,5),(4,8),(5,9)$.

1. Draw the scatter plot and describe direction, form and apparent strength.
2. Calculate $\bar{x}$, $\bar{y}$, $V(x)$, $V(y)$, $\mathrm{Cov}(x,y)$ and correlation r.
3. Estimate the least-squares line $\hat{y} = ax + b$.
4. Calculate fitted values and residuals $e_i = y_i - \hat{y}_i$.
5. Calculate SSE, SST and R^2 and interpret R^2.

Assessment / exam practice — 15 min

2.2 Exercise 4 – Two-stage forecast

Month	1	2	3	4	5	6	7	8
Advertising X	12	15	14	18	20	22	25	27
Sales Y	48	52	50	59	63	68	73	78

Given $\widehat{Y} = 2.064X + 21.899$ and $\widehat{X}_t = 2.155t + 9.429$, forecast month-10 advertising and sales, and identify where extrapolation occurs.

Assessment / exam practice — 45 min

2.3 Exercise 5 — Two successive regressions — exam style

For six periods, observe period t, advertising x and sales y.

t	1	2	3	4	5	6
x	4	5	7	8	10	11
y	10	12	15	18	22	24

1. Build the complete t, x, y, t^2, x^2, y^2, tx, xy table and sums.
2. Compute means, variances, covariance, r and the regression of x on t.
3. Repeat for y on x.
4. Forecast x then y for t=8; discuss extrapolation.

3 Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice — 20 min

3.1 Exercise 2 – Interpolation, extrapolation, causality and influence

Using Exercise 1, forecast Y at $X = 3.5$ and $X = 10$, classify each forecast, add point $(10, 30)$ and recalculate the regression line, then explain why high correlation does not establish causality.

Advanced practice — 20 min

3.2 Extra 1 — Regression diagnosis and responsible decision

A model linking store traffic to temperature reports r = 0.78 and $R^2 = 0.61$. Its residual plot is U-shaped, one summer weekend has a very large residual, and the manager wants to forecast a temperature 12řC above the observed maximum.

1. Interpret r and R^2 correctly.
2. Diagnose the residual pattern.
3. Discuss the unusual weekend and the proposed forecast.
4. Recommend the next analytical steps.

Method hint:

Separate strength, functional form, unusual observations, extrapolation and causality.